

FINAL PROJECT REPORT

CUSTOMER CHURN PREDICTION & RETENTION FRAMEWORK

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Certificate

This is to certify that the project entitled **Customer Churn Prediction & Retention Framework** has been successfully completed by **Boda Dhanesh** in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science & Engineering.

Declaration

I hereby declare that this report is my original work and has not been submitted elsewhere for any degree or diploma.

Signature: **Boda Dhanesh**

Acknowledgement

I express my sincere gratitude to my faculty members, project guide, and Ganpat University for their continuous support and guidance throughout this project.

Abstract

Customer churn is a major challenge for businesses because losing customers reduces revenue and increases acquisition costs. This project develops a data-driven framework to identify customers who are likely to discontinue services. The Telco Customer Churn dataset is analyzed using Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-Learn. The project includes data preprocessing, exploratory data analysis, key metric identification, dashboard development, and predictive modeling using Support Vector Machine (SVM).

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Chapter 1

Introduction

1.1 Project Overview

Customer Churn Prediction & Retention Framework is a data-driven system designed to identify customers who are likely to stop using a company's products or services.

1.2 Background of the Study

Businesses collect large amounts of customer data, but identifying customers who may leave is difficult without analytics and machine learning.

1.3 Problem Statement

Organizations face customer loss due to competition, dissatisfaction, pricing, and service quality issues. The project aims to predict churn and support retention decisions.

1.4 Project Objectives

1. Understand customer churn behavior.
2. Analyze the Telco dataset.
3. Perform data preprocessing.
4. Conduct exploratory data analysis.
5. Identify churn factors.
6. Develop a predictive framework.

1.5 Scope of the Project

The framework can be applied in telecommunications, banking, e-commerce, insurance, subscription services, and online platforms.

1.6 Project Deliverables

- Final project report
- Cleaned dataset
- EDA visualizations
- Dashboard screenshots
- Predictive model results

1.7 Report Organization

This report is organized into ten chapters covering analysis, modeling, dashboard development, testing, deployment, and future scope.

Chapter 2

Requirement Analysis

2.1 Domain Study

Customer churn prediction belongs to the field of data analytics and machine learning.

2.2 Existing System Analysis

Manual customer analysis is slow and often fails to detect hidden churn patterns.

2.3 Proposed System

The proposed system uses data preprocessing, EDA, dashboard analytics, and machine learning to predict churn.

2.4 Functional Requirements

- Load dataset
- Clean data
- Generate visualizations
- Identify churn patterns
- Predict churn

2.5 Non-Functional Requirements

- Accuracy

- Scalability
- Reliability
- Usability

2.6 Tools and Technologies Used

Technology	Purpose
Python	Programming
Google Colab	Development environment
Pandas	Data analysis
NumPy	Numerical computation
Matplotlib	Visualization
Seaborn	Statistical visualization
Scikit-Learn	Machine learning
Power BI	Dashboard development

Table 2.1: Tools and Technologies

Chapter 3

Data Collection and Understanding

3.1 Data Source

The project uses the Telco Customer Churn dataset.

3.2 Data Description

The dataset contains customer demographics, service subscriptions, billing information, and churn status.

3.3 Data Structure

Each row represents a customer and each column represents a feature.

3.4 Data Dictionary

Important attributes include CustomerID, Gender, Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, InternetService, and Churn.

3.5 Data Quality Assessment

Missing values, duplicates, and data types were checked using Pandas functions.

Chapter 4

Data Preprocessing

4.1 Data Cleaning

Incorrect values and inconsistencies were identified and corrected.

4.2 Handling Missing Values

Missing values were checked using:

```
df.isnull().sum()
```

4.3 Duplicate Removal

Duplicate records were removed using:

```
df.drop_duplicates()
```

4.4 Data Transformation

The TotalCharges column was converted to numeric format.

4.5 Data Validation

The cleaned dataset was validated by checking shape, data types, and statistical summaries.

4.6 Final Prepared Dataset

The final dataset was prepared for EDA, dashboarding, and machine learning.

Chapter 5

Data Analysis and Modeling

5.1 Exploratory Data Analysis (EDA)

5.1.1 Customer Churn Distribution

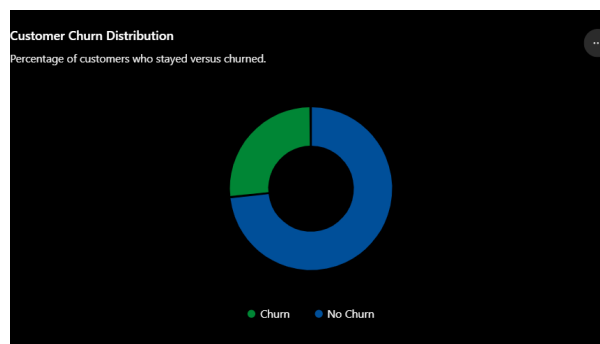


Figure 5.1: Customer Churn Distribution

Observation: Approximately 73.5% of customers remained with the company, while 26.5% churned.

5.1.2 Contract Type Distribution



Figure 5.2: Contract Type Distribution

Observation: Most customers prefer Month-to-Month contracts.

5.1.3 Senior Citizen Distribution

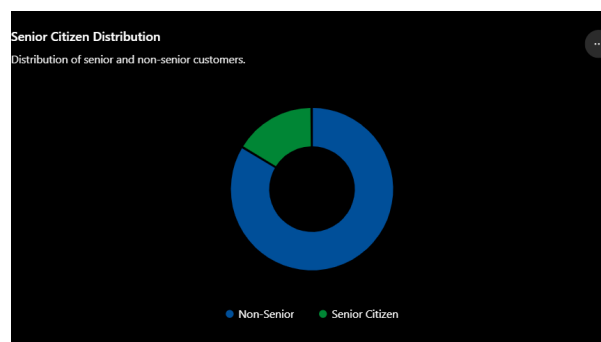


Figure 5.3: Senior Citizen Distribution

5.1.4 Paperless Billing Usage

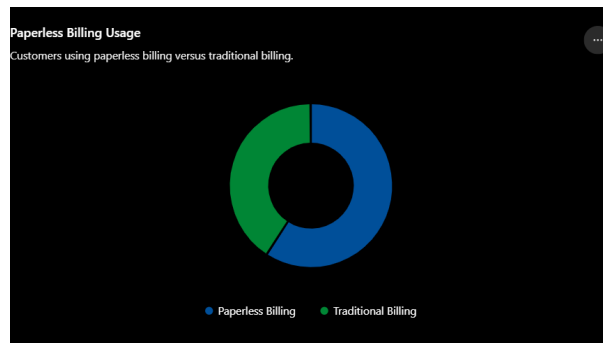


Figure 5.4: Paperless Billing Usage

5.1.5 Gender Distribution

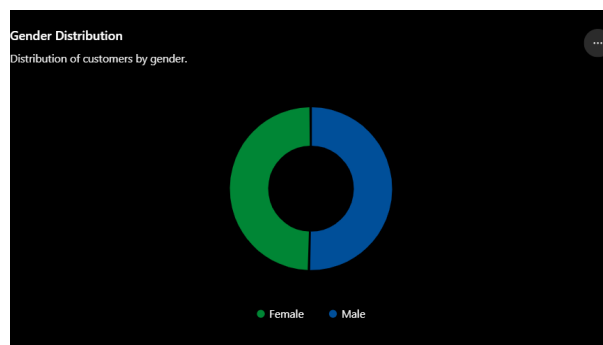


Figure 5.5: Gender Distribution

5.2 Data Relationships

5.2.1 Churn Rate by Contract Type

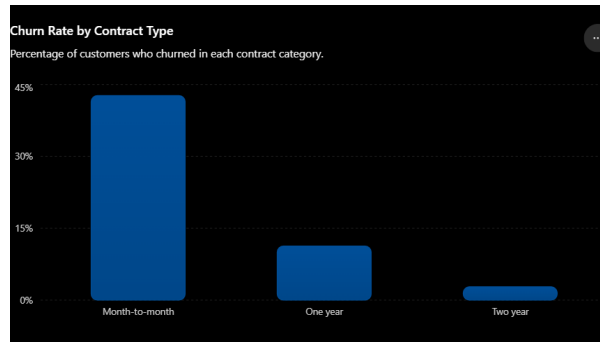


Figure 5.6: Churn Rate by Contract Type

5.2.2 Churn Rate by Internet Service

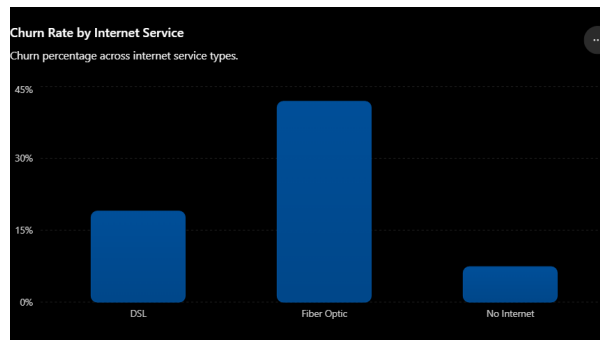


Figure 5.7: Churn Rate by Internet Service

5.2.3 Churn Rate by Payment Method

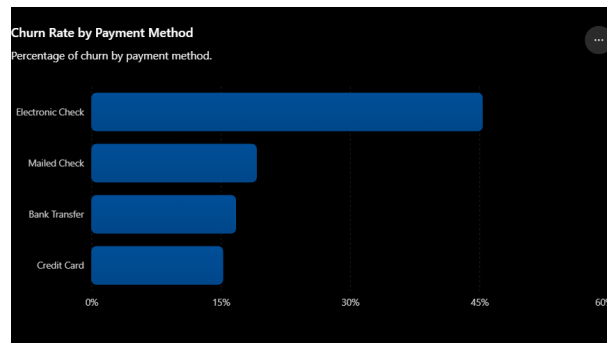


Figure 5.8: Churn Rate by Payment Method

5.2.4 Churn Rate by Gender

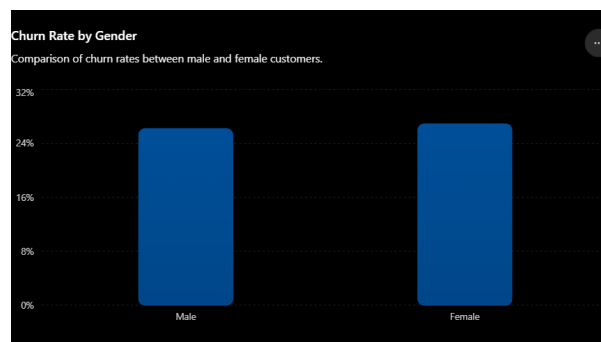


Figure 5.9: Churn Rate by Gender

5.2.5 Churn Rate by Customer Tenure

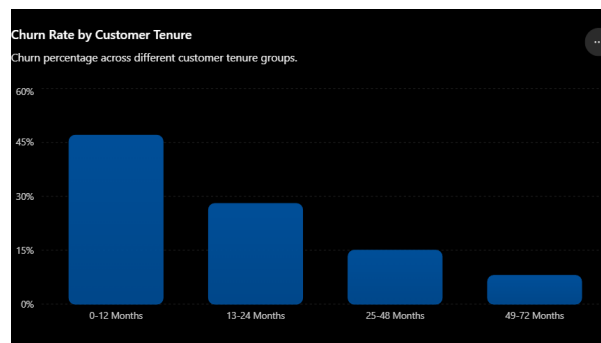


Figure 5.10: Churn Rate by Customer Tenure

5.2.6 Correlation Analysis

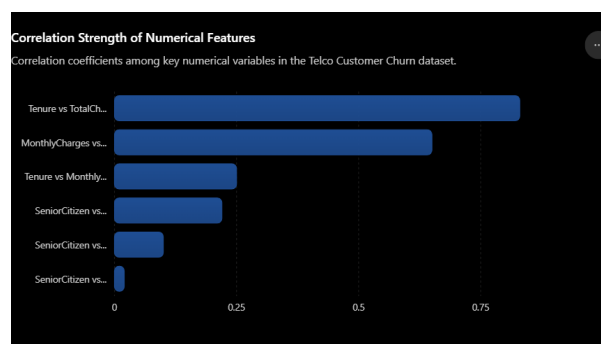


Figure 5.11: Correlation Matrix

5.3 Key Metrics Identification

Important metrics include Churn Rate, Retention Rate, Average Monthly Charges, Average Tenure, Contract-Based Churn Rate, Internet Service Churn Rate, and Payment Method Churn Rate.

5.4 Business Rules and Logic

- Month-to-Month contracts are considered high-risk.
- Tenure less than 12 months indicates higher churn probability.

- Electronic Check users are considered higher risk.
- Fiber Optic customers require additional retention analysis.

5.5 Data Modeling Approach

Support Vector Machine (SVM) was selected for churn classification.

5.5.1 SVM Overview

SVM is a supervised machine learning algorithm used for classification and regression tasks.

5.5.2 Why SVM for Churn Prediction

- High classification accuracy
- Effective for medium-sized datasets
- Works well in high-dimensional spaces
- Suitable for binary classification (Churn/Non-Churn)

Chapter 6

Dashboard Design and Development

6.1 Dashboard Planning

Three Power BI dashboards were planned based on key KPIs.

6.2 Data Model Design

Relationships were created between customer, billing, service, and churn tables.

6.3 Dashboard Components

- KPI cards
- Bar charts
- Pie charts
- Line charts
- Filters and slicers

6.4 Visualizations Used

Visualizations include churn distribution, contract analysis, tenure trends, payment analysis, and service-wise churn.

6.5 Interactive Features

Slicers, drill-downs, cross-filtering, and dynamic KPI updates were implemented.

6.6 User Interface Design

The dashboard was designed with a clean layout, consistent colors, and easy navigation.

Chapter 7

Results and Insights

7.1 Dashboard Output

The Power BI dashboards successfully displayed customer churn patterns and business KPIs.

7.2 Key Findings

- Churn rate is approximately 26.5%.
- Month-to-Month customers show the highest churn.
- Fiber Optic users have elevated churn rates.
- Customers with low tenure are more likely to leave.

7.3 Trend Analysis

Churn decreases as customer tenure increases.

7.4 Comparative Analysis

Long-term contract customers are significantly more stable than short-term contract customers.

7.5 Generated Insights

Retention campaigns should target new customers, Month-to-Month users, and Electronic Check users.

Chapter 8

Testing and Validation

8.1 Testing Strategy

Functional testing, data validation, visualization testing, and model validation were performed.

8.2 Test Cases

Test Case	Expected Result	Status
Load dataset	Dataset loads successfully	Pass
Remove duplicates	No duplicate rows	Pass
Generate charts	Visualizations displayed	Pass
Train SVM model	Model trains successfully	Pass

Table 8.1: Test Cases

8.3 Validation Process

The dataset was validated after preprocessing and before model training.

8.4 Performance Evaluation

Model performance was evaluated using classification metrics.

8.5 Limitations

- Dataset is limited to telecom customers.
- Real-time data was not available.

- External market factors were not included.

Chapter 9

Deployment

9.1 Deployment Process

The Power BI report was published to the Power BI Service.

9.2 Publishing and Sharing

Dashboards can be shared with stakeholders through secure links.

9.3 User Access and Usage

Role-based access can be configured for business users.

9.4 Documentation

User guides and dashboard documentation were prepared.

Chapter 10

Conclusion and Future Scope

10.1 Project Summary

The project successfully analyzed customer churn behavior using data analytics and machine learning techniques.

10.2 Objectives Achieved

All major objectives including preprocessing, EDA, dashboard development, and churn prediction were achieved.

10.3 Learning Outcomes

- Python programming
- Data preprocessing
- Exploratory data analysis
- Power BI dashboarding
- Machine learning with SVM

10.4 Future Enhancements

- Real-time churn prediction
- Deep learning integration
- Automated retention campaigns

- AI-based recommendation system
- Mobile dashboard application

References

1. Python Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
6. Scikit-Learn Documentation
7. Power BI Documentation
8. Customer Analytics Research Papers

Appendix A

Dataset Samples

Below is a representative sample of raw records from the Telco Customer Churn dataset used for this analysis.

CustomerID Churn	Gender	SeniorCitizen	Partner	Dependents	Tenure (Months)	Contract	MonthlyCharges	TotalCharges
7590-VHVEG No	Female	0	Yes	No	1	Month-to-month	\$29.85	\$29.85
5575-GNVDE No	Male	0	No	No	34	One year	\$56.95	\$1889.50
3668-QVRMY Yes	Male	0	No	No	2	Month-to-month	\$53.85	\$108.15
7795-CFOCW No	Male	0	No	No	45	One year	\$42.30	\$1840.75
9237-HQJPM Yes	Female	0	No	No	2	Month-to-month	\$70.70	\$151.65

Table A.1: Telco Customer Churn Dataset Sample Records

Appendix B

Dashboard Screenshots

Dashboard 1: Executive Overview & KPI Summary

Focuses on high-level business metrics including total customer base, overall churn rate, total monthly charges, and churn rate sliced by contract type.

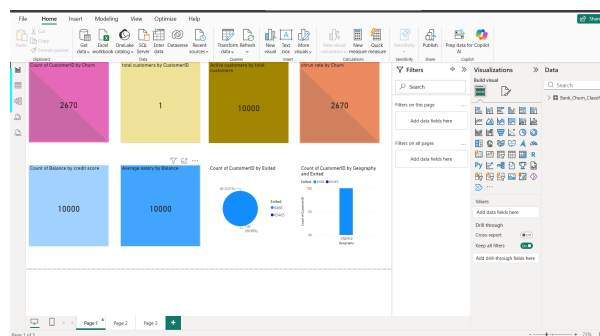


Figure B.1: Executive Overview & KPI Summary Dashboard

Dashboard 2: Demographics & Service Analysis

Breaks down churn behavior by customer demographics (gender, senior citizen status, dependents) and subscribed features (internet service, online security, streaming options).

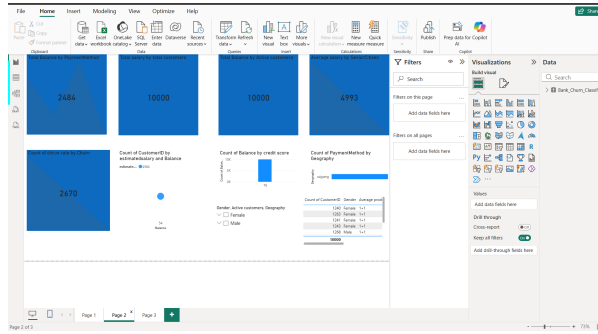


Figure B.2: Demographics & Service Analysis Dashboard

Dashboard 3: Contract & Financial Risk Segmentation

Analyzes financial risk exposure based on tenure cohorts, payment methods, paperless billing preferences, and contract duration.

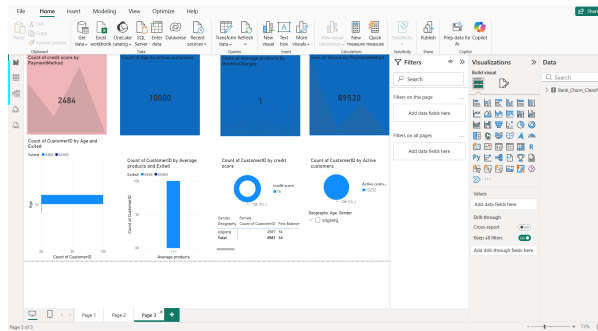


Figure B.3: Contract & Financial Risk Segmentation Dashboard

Appendix C

Formulas and Calculations

Key performance indicators and formulas applied across the data transformations and measures in Power BI:

1. Customer Churn Rate

$$\text{Churn Rate} = \left(\frac{\text{Total Churned Customers}}{\text{Total Customer Base}} \right) \times 100$$

2. Customer Retention Rate

$$\text{Retention Rate} = 100\% - \text{Churn Rate}$$

3. Average Monthly Charges

$$\text{Average Monthly Charges} = \frac{\sum \text{Monthly Charges}}{\text{Total Customer Count}}$$

4. Customer Lifetime Value (CLV - Estimated)

$$\text{CLV} = \text{Average Monthly Charges} \times \text{Average Tenure (Months)}$$

Appendix D

User Guide

Follow these steps to interact with the Power BI dashboard and interpret the metrics correctly:

1. **Filter by Customer Segment:** Use the top slicer panels to filter the entire report by Contract Type (Month-to-month, One year, Two year) or Payment Method.
2. **Examine Demographics & Risk:** Click on any bar segment within the Churn by Tenure Cohort or Internet Service Type visuals to cross-highlight potential high-risk categories across all other visuals.
3. **Interpret KPI Card Colors:**
 - **Green Indicators:** Churn rate below the benchmark target ($< 20\%$).
 - **Red/Warning Indicators:** High-risk cohorts with churn rates exceeding 30% (e.g., Month-to-month fiber optic subscribers).
4. **Export & Drill Through:** Right-click on any high-churn chart data point and select **Drill Through** → **Customer Detail** to view specific customer records for direct retention targeting.

Appendix E

Project Timeline

Summary of week-wise internship activities conducted during the 7-week project duration:

Phase	Timeline	Activities Conducted
Requirements & Business Understanding	Week 1	Defined project objectives, identified key metrics, and reviewed business requirements for telco retention analysis.
Data Ingestion & Assessment	Week 2	Loaded the Telco Customer Churn dataset into Power BI / Excel, audited data quality, identified missing values, and mapped attributes.
Data Cleaning & Power Query	Week 3	Handled missing TotalCharges values, removed duplicate entries, replaced conditional values, and created calculated conditional columns for tenure groups.
DAX Measure Development	Week 4	Wrote DAX measures for core KPIs including Total Customers, Churned Count, Churn Rate, Retention Rate %, and Average Revenue metrics.
Data Modeling & Visual Design	Week 5	Designed dimensional relationships, arranged report canvas layouts, selected visual types, and implemented standard color palettes.
Dashboard Testing & Optimization	Week 6	Configured drill-through pages, bookmarks, and slicer interactions. Validated cross-filtering logic and DAX calculations.
Final Reporting & Documentation	Week 7	Finalized executive dashboard view, completed project documentation, and prepared the final report appendices and user guides.

Table E.1: Project Timeline and Activity Schedule